# Stealthy Adversarial Examples for Semantic Segmentation in Remote Sensing

Tao Bai, Yiming Cao, Yonghao Xu, *Member, IEEE*, and Bihan Wen, *Senior Member, IEEE*

***Abstract*— Deep learning methods have been proven effective in remote sensing image analysis and interpretation, where semantic segmentation plays a vital role. These deep segmentation methods are susceptible to adversarial attacks, while most of the existing attack methods tend to manipulate the image globally, leading to noticeable perturbations and chaotic segmentation. In this work, we propose a novel stealthy attack for semantic segmentation (SASS), which can largely increase the effectiveness and stealthiness from the existing attack methods on remote sensing images. SASS manipulates specific victim classes or objects of interest while preserving the original segmentation results for other classes or objects. In practice, as different inference mechanisms, overlapped inference, can be applied in segmentation, the efficacy of SASS may be degraded. To this end, we further introduce the masked SASS (MSASS), which generates augmented adversarial perturbations that only affect victim areas. We evaluate the effectiveness of SASS and MSASS using four state-of-the-art semantic segmentation models on the Vaihingen and Zurich Summer datasets. Extensive experiments demonstrate that our SASS and MSASS methods achieve superior attack performances on victim areas while maintaining high accuracies of other areas (drop less than 2%). The detection success rates of adversarial examples for segmentation, as characterized by Xiao et al., significantly drop from 97.78% for the untargeted projected gradient descent (PGD) attack to 28.71% for our MSASS method on the Zurich Summer dataset. Our work contributes to the field of adversarial attacks in semantic segmentation for remote sensing images by improving stealthiness, flexibility, and robustness. We anticipate that our findings will inspire the development of defense methods to enhance the security and reliability of semantic segmentation models against our stealthy attack.**

***Index Terms*— Adversarial attack, deep learning, remote sensing, semantic segmentation.**

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## I. Introduction

SEMANTIC segmentation is a fundamental task in the analysis and interpretation of the diverse and data-rich remote sensing images [2] and has been applied in various important applications, including urban land use analysis [3] and environmental monitoring [4]. In the last few years, with the tremendous success of deep neural networks, many deep learning architectures, such as convolutional neural networks (CNNs), have been applied to remote sensing [5], [6], [7], demonstrating the superior results in semantic segmentation of remote sensing imagery.

On the other hand, deep neural networks are known to be highly vulnerable to adversarial attacks [8], [9], [10], [11], i.e., the input with specifically crafted perturbations to undermine the accuracy and reliability of the targeted artificial intelligence (AI) systems. In remote sensing, concerns regarding the security of semantic segmentation models against adversarial examples have gained great attention [12]. While some research in computer vision has focused on adversarial attacks in semantic segmentation [13], [14], these methods primarily aim to disrupt the overall predictions by assigning wrong predictions to each pixel, paying limited attention to the stealthiness of results after attacks .[1] As shown in Fig. 1, the attacked image exhibits disordered and unpredictable deviations from its original segmentation results. For instance, part of the "Water" area is mixed up with either "Buildings" or "Bare soil" region, which easily raises suspicions regarding the reliability of the predictions. Consequently, such attacks can be easily detected by human or using defense mechanisms such as spatial consistency analysis [1].

The focus on deep learning model vulnerability in remote sensing is prominent in image classification and object detection, but lacking in semantic segmentation, warranting investigation. The complexity of interpixel relationships compounds the challenge of executing effective attacks in this context [12]. Unlike conventional methods targeting broad misclassification, our study is motivated by the unique characteristics of remote sensing imagery—high resolution, substantial size, and rich content—creating an ideal environment for stealthy yet impactful attacks. For instance, the research explores attacks on specific entities such as buildings

[1]Traditional adversarial attacks only focused on minimizing the adversarial perturbation budget. Following the same stealthy idea defined in [15], here we emphasize the unnoticeableness of prediction results after attacks.

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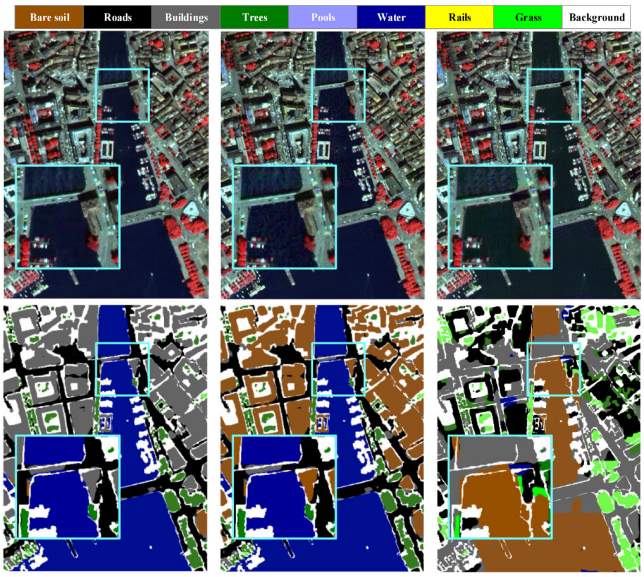



Fig. 1. Comparison between PGD and SASS attack. (a) Original clean test image and segmentation results of ICNet in the Zurich Summer dataset. (b) Adversarial test image and segmentation result after "Buildings → Bare soil" SASS attack. Segmentation results of ICNet on the clean images. (c) Adversarial test image and segmentation result after PGD attack. The PGD attack generates completely disordered predictions. The SASS attack manipulates pixels to be classified as "Bare soil" instead of "Buildings" while preserving the contextual information of the remaining pixels, thereby making SASS not easy to spot.

or vehicles, redirecting their classification while meticulously preserving other prediction accuracies.

In the realm of remote sensing semantic segmentation, where images encompass diverse objects and scenes, some of which hold critical implications for security, an attacker may seek to target one specific object, such as one specific building or car. This involves the intent to secretly reclassify it into another class, such as low vegetation or road. Achieving this objective necessitates precision to avoid misclassifying other nontarget classes in the prediction results to avoid suspicion. The intricacy arises from the interdependence of every pixel in semantic segmentation, making the execution of the nuanced attack challenging. Given that not all areas in remote sensing images are relevant or of interest in real-world scenarios, it becomes less meaningful for attackers to manipulate the entire image and generate completely different segmentation results. Since those chaotic segmentation results are likely to increase suspicion of human and be noticed, such attacks would be defended in time and end with failure. Instead of pursuing a global objective of misleading all predictions made by segmentation models, a more refined and stealthier attack goal [16] is introduced in this article and demonstrated in Fig. 1. We seek to target at specific victim source classes or areas of interest and manipulate them to be classified as another target class while ensuring that pixels belonging to nonvictim classes or areas remain unaffected by the attack. Thus, the overall visual coherence of the image can be maintained and the risk of being detected can be reduced. To this end, we propose stealthy attack for semantic segmentation (SASS), which aims to achieve both stealthiness and strong attack abilities on remote sensing segmentation. In SASS, the perturbations are strategically produced to manipulate specific victim classes or areas and overall disruptions of the whole image are imperceptible.

Due to the high resolution of remote sensing images, remote sensing images are usually cropped into small patches to fit the memory when feeding into deep learning models, which is different from segmentation in computer vision. Moreover, overlapped inference [7], [17], where input patches are cropped with overlaps, is also a common technique to avoid the border inconsistent phenomenon and improve models' performances. These two operations would introduce additional transformations to the inputs and break the adversarial patterns generated by SASS, which is also consistent with our experimental observations. Recognizing the importance of addressing this limitation, we introduce the masked SASS (MSASS) method, where the augmented perturbations are only added to the victim areas of interest and resistant to overlapped inference. In summary, the contributions of this article are threefold.

1) To the best of our knowledge, we are the first to study the stealthiness of adversarial examples for semantic segmentation in remote sensing and aim to generate effective and stealthy adversarial examples.
2) We propose SASS, which enables adversarial attacks of semantic segmentation to achieve a higher level of stealthiness. Through extensive experiments on the Vaihingen and Zurich Summer datasets, SASS shows strong attack abilities against various state-of-the-art semantic segmentation models.
3) We further propose MSASS with practical considerations and provide flexible control over the attack with specified victim classes and areas. In addition to strong attack abilities, MSASS results in a great drop in the success rates of adversarial detection methods.

The following of this article is organized as follows. Section II provides a literature review, discussing semantic segmentation models in remote sensing and adversarial examples against semantic segmentation. In Section III, we present the details of our proposed SASS and MSASS methods. Section IV focuses on our experiments, including the dataset, experimental settings, and a comprehensive analysis of the results. Finally, Section V concludes this article.

## II. Related Works

### A. AI for Remote Sensing

Remote sensing employs sensors detecting diverse forms of energy emitted, reflected, or scattered by the target object, enabling data acquisition without direct physical interaction [18]. The pervasive use of sensor technologies has significantly augmented remote sensing data volume, providing a robust foundation for integrating AI methodologies. AI methodologies have been applied across domains of remote sensing including low-level vision tasks such as pan sharpening [19] and cloud removal [20], high-level vision tasks such as object detection [21] and semantic segmentation [22], and environment monitoring tasks such as wildfire detection [23] and weather forecasting [24].

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Conventional machine learning approaches, such as ensemble decision-tree-based classifiers and support vector machines (SVMs), demonstrate resilience to noisy data and possess the capability to manage the high dimensionality inherent in remote sensing datasets, which are often characterized by numerous spectral bands and variables [25]. These methods find application across various remote sensing tasks, including tree species classification [26] and land-cover classification [27], [28], [29], [30]. For instance, Yang et al. [29] developed a machine learning model based on the random forest algorithm to generate snow-covered area maps from PlanetScope imagery at a spatial resolution of approximately 3 m. In recent years, deep learning has emerged as a valuable tool in remote sensing, providing effective solutions across various tasks, including preprocessing [20], [31], object recognition [32], semantic segmentation [22], [33], and environmental monitoring [23]. Remote sensing data often exhibit spatial and temporal dependencies. Deep learning architectures, such as convolutional neural networks (CNNs) designed for spatial data and recurrent neural networks tailored for temporal data, have proven effective in addressing these dependencies [24]. Furthermore, attention mechanisms have played a crucial role in enhancing the model performance in remote sensing by dynamically focusing on specific regions or features within input data, leading to more accurate results. For instance, Zhao et al. [20] employed a multimodal attention mechanism with extracted and aligned radar and multispectral images to address the challenge of cloud removal in remote sensing. Zhang et al. [21] fused multimodal data and introduced assisted super-resolution learning into high-resolution object detection on multiscale objects, achieving high detection accuracy and inference speed. Rashkovetsky et al. [23] employed a fully convolutional network (FCN) for deep semantic segmentation in detecting wildfires from single-temporal data obtained across the visible, infrared, and microwave domains.

### B. Adversarial Examples of Remote Sensing Images

Given the safety-critical nature of many remote sensing tasks, it is imperative to address inherent vulnerabilities and uncertainties in AI models [34], such as the risk of adversarial attacks, which involve introducing specifically crafted perturbations to compromise the accuracy and reliability of targeted AI systems [8], [9], [10], [11]. In the context of adversarial attacks against image classification, Czaja et al. [35] investigate adversarial examples in the context of satellite image classification and outline a method for simulating physical attacks on remotely sensed data. Chen et al. [36] also assess adversarial examples on remote sensing image scene classification, considering both optical and synthetic aperture radar (SAR) images with varying spectral and spatial resolutions. Research on adversarial patch attacks for optical remote sensing images has also been explored [37], [38], [39]. These attack methods typically involve generating an adversarial patch to attack an object detector. Sun et al. [39] propose threatening patch attacks (TPAs) on object detection in optical remote sensing images that also pose the attack to local image regions. Our work, represented by SASS and MSASS, diverges from theirs in three key aspects. First, TPA's attack on local regions lacks the precision and continuity observed in our approach. TPA computes the attack on a selected subpatch within the bounding box of an object instance, in contrast to our pixel-level calculation. Moreover, TPA generates an attack map, but the resulting adversarial perturbation affects the entire image rather than just the targeted instance. In contrast, our masked attack confines the perturbation to the region corresponding to the object of interest. Finally, our targeted attack strategy exhibits a nuanced objective: to manipulate the object's classification from a source class to a target class while maintaining accurate predictions for nonsource pixels. Conversely, TPA focuses on misleading either the classification or regression branch without the subtlety inherent in our targeted approach. Few works have focused on adversarial attacks against semantic segmentation in remote sensing. Xu and Ghamisi [12] explore the universal adversarial examples in remote sensing data introducing the Mixup-Attack method for an untargeted black-box adversarial attack on remote sensing scene classification and semantic segmentation. Since their attack is untargeted and focuses on black-box attacks, thus it is unsuitable for comparison with our proposed method. Na et al. [40] propose segmentation networks based on a domain adaptive transfer attack (DATA) scheme for building extraction from aerial images. The DATA scheme seeks to align input image distribution with the target images, transforming them into adversarial examples against a target network. While enhancing model robustness, its untargeted nature and specific design for building extraction from aerial images make it inappropriate for comparison with our proposed method.

### C. Semantic Segmentation

Semantic segmentation plays a crucial role in computer vision task that involves assigning a semantic label to each pixel in an input image. Initially, random forest and conditional random field (CRF) methods are used for semantic segmentation [41]. Nevertheless, deep learning has emerged as a prominent alternative to traditional methods. In recent years, the increasing availability of satellite and airborne sensors has led to a surge in diverse and data-rich remote sensing imagery, capturing vast areas of the Earth's surface [42]. Consequently, semantic segmentation in remote sensing garners significant research attention and plays a fundamental role in various applications, including environmental monitoring, building classification, and land use analysis [2]. Accurate and efficient semantic segmentation of remote sensing imagery provides valuable insights for understanding and managing the Earth's resources.

One classical model for semantic segmentation is the FCN [43], which can process inputs of any size and generate the corresponding outputs through pixels-to-pixels training. Taking into account the high resolution of aerial images and memory limitations [7], [17] in the domain of remote sensing, researchers have proposed the use of patch-based CNNs to predict classification maps. In particular, Pan et al. [7] extend

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the densely connected convolution with pyramid pooling modules to effectively fuse multiresolution features extracted from high-resolution aerial imagery. Ronneberger et al. [44] introduce the U-Net architecture for biomedical image segmentation. The U-Net comprises a contracting path that captures contextual information and an expanding path that enhances localization accuracy. Building upon this work, Diakogiannis et al. [5] extend the U-Net framework for semantic segmentation of remote sensing data. Specifically, they proposed ResUNet-a, which leverages a U-Net encoder/decoder backbone along with various enhancements and demonstrates competitive performance. These enhancements include residual connections, atrous convolutions, pyramid scene parsing pooling, and multitasking inference. DeepLabV3+ [45] is another widely used model for semantic segmentation. It is an improvement over DeepLabV3 [46], which uses a spatial pyramid pooling module to encode multiscale contextual information with an encoder–decoder structure to recover the spatial information gradually. DeepLabV3+ improves DeepLabV3 with an efficient decoder module, further refining the segmentation results, especially in recovering detailed object boundaries. Arief et al. [6] employ the DeepLab model to perform land-cover segmentation of airborne light detection and ranging (LiDAR) data. They proposed a fusion approach that combines spatial features with height above ground information and intensity values, tailored to the specific requirements of the task. Real-time semantic segmentation is a popular task since it considers the tradeoff between accuracy and inference speed under applications with resource constraints [41]. One model that has gained popularity in the field of real-time semantic segmentation is the ICNet [47]. ICNet processes the low-resolution inputs with a lightweight CNN to obtain content information and then fuses this information with the high-resolution inputs to obtain accurate pixel-level segmentation. Drawing inspiration from the framework of ICNet, He et al. [48] propose MFFANet to perform semantic segmentation of remote sensing images, which introduces a multiscale attention refinement module. This module aggregates information from both high- and low-level features, simultaneously restoring image resolution to improve segmentation performance.

CNN-based methodologies encounter challenges in effectively capturing a comprehensive global context due to the inherent local nature of the convolution operation. Recent advancements in remote sensing segmentation highlight the increasing importance of employing transformer-based approaches. He et al. [33] propose a dual encoder structure by integrating the Swin transformer [49] into the U-Net framework in parallel. Wang et al. [22] adopt a hybrid architecture, connecting a CNN-based encoder with a transformer-based decoder. To improve the model's capacity for capturing global contexts with minimal complexity, they introduce window-based multihead self-attention and a cross-shaped window context interaction module.

### *D. Adversarial Examples Against Semantic Segmentation*

An adversarial example against semantic segmentation refers to a deliberately modified input image that attempts to deceive a segmentation model, resulting in incorrect segmentation output. According to recent research [12], semantic segmentation models are generally more resistant to adversarial attacks compared to models in domains such as image classification since semantic segmentation models require a more detailed understanding of the spatial relationship between pixels in an image, making it harder to generate perturbations that can cause significant changes in the output segmentation map.

There are generally two approaches to constructing adversarial examples against semantic segmentation. One is directly optimizing the perturbation according to the specific target loss [13], [14], [50], and another is training a perturbation generator [15]. Xie et al. [13] propose dense adversary generation (DAG) that iteratively suppresses the classification confidence of the correctly classified class and increases the classification confidence of a randomly selected wrong class of the adversarial example. Metzen et al. [14] propose targeted universal adversarial examples against semantic segmentation that remove a target class by minimizing the loss between the prediction and the adversarial target. The adversarial target is generated through a nearest neighbor heuristic. Gu et al. [50] refine the projected gradient descent (PGD) method by dynamically adjusting the weight of the loss generated from both correctly and wrongly classified pixels to generate more effective and efficient adversarial examples. Chen et al. [15] utilize a generator to generate the adversarial example and a regularizer to predict segmentation labels. The manipulated labels are paired with the predicted labels to calculate the adversarial loss. The authors propose targeted attack types aimed at achieving either the disappearance of the target object into the background or the incorporation of new pixels from an external source. However, these attacks exhibit limited flexibility, as they are unable to perform specific tasks without relying on external sources. Chen et al. [15] also propose to mitigate the impact on nontargeted labels. It is worth noting that their approach, while effective, exhibits limitations in terms of computational efficiency, due to the reliance on a CNN-based generator and regularizer. For remote sensing image semantic segmentation, Xu and Ghamisi [12] propose the universal black-box Mixup-Attack, which utilizes the common vulnerability of different networks and attacks the features of the shallow layer of the victim model. Wang et al. [51] propose global feature attention network to detect FGSM [8], C&W [10], and PGD [9] adversarial examples against aerial image semantic segmentation.

Although effective in misleading deep neural networks with imperceptible perturbations, adversarial examples also face several challenges. One of these challenges is the limited transferability of adversarial examples across different transformations. Lu et al. [52] find that adversarial examples do not transfer well across different transformations. Arnab et al. [53] find that adversarial examples in semantic segmentation undergoing transformations will not be as malicious since CNNs are inherently not invariant to transformations such as translation [54] and the activations of the network change greatly compared to the original input after transformations. Moreover, to address the vulnerabilities posed

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by adversarial examples, Xiao et al. [1] propose to detect adversarial examples against semantic segmentation through spatial consistency information. They also show that such spatial consistency analysis is effective even when facing adaptive attacks.

## III. Methodology

### A. Preliminaries

In semantic segmentation, let $X \in R^{h\times w\times c}$ be an image that contains $h \times w$ pixels and $c$ channels. Each pixel $p \in \{X_{i,j} : 1 \le i \le h, 1 \le j \le w\}$ is assigned a ground-truth class label $y$, where $p \in R^c$ and $y \in R$. The pixel-level classification function of the victim model for the pixel $p$ in image $X$ is denoted as $f(X, p)$. Here, the victim model corresponds to the semantic segmentation model under consideration for the attack. The general objective of an adversarial attack is to create an adversarial example $X_{\text{adv}}$ to mislead classifications of all pixels of an input image [50], which is expressed as follows:

$$\begin{aligned} &f(X_{\text{adv}}, p) \neq f(X, p), \text{ where} \\ &X_{\text{adv}} = X + \delta \text{ subject to } \|\delta\|_\infty \le \epsilon \end{aligned} \tag{1}$$

where $\epsilon$ is the maximum magnitude of perturbations $\delta$ under $l_\infty$ norm.

Such attacks disrupt all predictions of segmentation models on input images and lead to disordered and unpredictable segmentation results. In real-world scenarios, however, desirable attacks are those working as expected but without raising alarms for special purposes, such as camouflaging a car as part of the road or transforming a vegetated area into a fictitious building. Conventional attacks apparently are not suitable in such scenarios. As such, we propose two stealthy attacks: SASS and MSASS that well fulfill the above attack goals.

### B. Stealthy Attack for Semantic Segmentation

Different from the general attacks, our SASS (illustrated in Fig. 2) specifies the victim source class $y_s$ and the target class $y_t$. Note that the target class should not be the same as the source class. The goal of SASS is to manipulate pixels from the source class to be classified as the target class while ensuring that nonsource class areas remain unaffected, which is formally formulated as follows:

$$\begin{aligned} f(X_{\text{adv}}, p) &= \begin{cases} y_t & \text{if } f(X, p) = y_s \\ f(X, p) & \text{otherwise} \end{cases} \\ X_{\text{adv}} &= X + \delta, \text{ subject to } \|\delta\|_\infty \le \epsilon. \end{aligned} \tag{2}$$

Correspondingly, we design a novel loss function denoted as $L_{\text{SASS}}$, comprising two components: $L_{\text{logits}}$ and $L_{\text{ce}}$. Logit loss has been demonstrated to have stronger gradients than cross-entropy (CE) loss [55] in attacking one class to target class since the gradient of the CE loss with respect to the logit of the target class will monotonically decrease as the probability of the target class increases during attack optimization and the gradient tends to vanish. Thus, in our first stealthy attack goal, which aims to transition from the source class to the target class, we follow the logit loss concept, designing losses directly on the logits of the source class and the sink class to ensure stronger and continuous improvement in attack performance. For the second stealthy attack goal, which aims to preserve the clean prediction of all other nonsource pixels, we use CE loss for efficiency.

We use $L_{\text{logits}}$ to make sure pixels from the source class to be misclassified as the target class, which is formulated in (3). There are two parts in $L_{\text{logits}}$. Specifically, $L_s$ is designed for decreasing the probabilities of pixels being predicted from the source class $y_s$, while $L_t$ is designed for increasing the probabilities of pixels being predicted as target class $t$

$$\begin{aligned} L_{\text{logit}} &= L_s + L_t, \text{ with} \\ L_s &= \max\left(l_s(X_{\text{adv}}, p_s) - \max_{i\neq y_s}(l_i(X_{\text{adv}}, p_s)), 0\right) \\ L_t &= \max\left(\max_{i\neq y_t}(l_i(X_{\text{adv}}, p_s) - l_t(X_{\text{adv}}, p_s)), -d\right) \end{aligned} \tag{3}$$

where $l_i(X, p)$ is the predicted logit of class $i$ of pixel $p$ in image $X$ normalized by the softmax function and $d$ is a hyperparameter to adjust the dominance of the target logit. Achieving a shift away from source class dominance is relatively straightforward, as perturbations can be strategically applied to elevate the logits of the second-likely class. Consequently, the threshold of $L_s$ is set to 0. However, establishing dominance for the randomly selected target class proves to be a more formidable challenge. Therefore, a concerted effort is required to increase the logits of the target class above all other possible classes, necessitating the introduction of the hyperparameter $d$ to exercise greater control over the enhancement of target class logits. In the study by Sun et al. [56], the balanced logit loss is proposed to suppress source logits while augmenting target logits. However, our approach differs significantly on two fronts. First, our focus lies on the semantic segmentation task, a pixel-level classification involving intricate interpixel spatial relationships. This task is empirically more challenging than the image classification task discussed in [56]. Second, we incorporate the consideration of the maximum prediction logit of source pixels. This addition ensures not only the superiority of target class logits over the source class but also the highest logit among all classes, more strictly fostering the target class to emerge as the predominant logit among all potential classes.

$L_{\text{ce}}$ is used to maintain the accuracy of the original predictions of pixels from nonsource classes, which is formulated as follows:

$$\begin{aligned} L_{\text{ce}} &= \sum_p \mathcal{X}(f(X_{\text{adv}}, p), \tilde{y}) \\ \tilde{y} &= \begin{cases} y_t & \text{if } f(X, p) = y_s \\ f(X, p) & \text{otherwise,} \end{cases} \end{aligned} \tag{4}$$

where $\mathcal{X}(\cdot)$ denotes the CE function and $\tilde{y}$ denotes the manipulated label. If the pixel is from the source class, the manipulated label $\tilde{y}$ is flipped to the target class $y_t$ and otherwise unchanged.

Finally, the overall loss function $L_{\text{SASS}}$ is formulated as follows:

$$L_{\text{SASS}} = (1-\alpha)L_{\text{logit}} + \alpha L_{\text{ce}} \tag{5}$$

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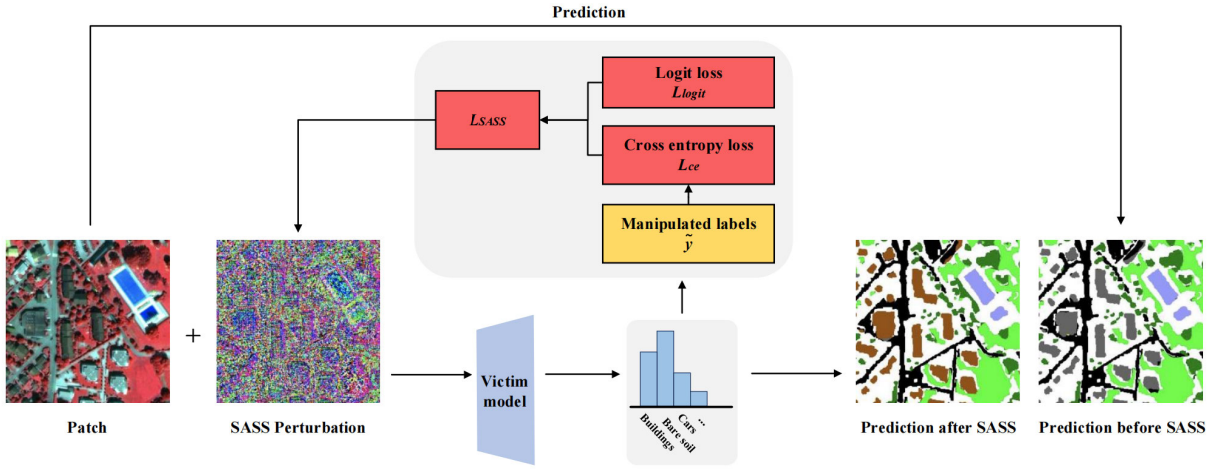



Fig. 2. Overview of our proposed SASS model. During training, the perturbation is added to the original image to produce an adversarial example. The pretrained victim model receives the adversarial image, and the predicted logits are acquired for optimizing the perturbation based on the attack scenario, where the source class is "Buildings" and the target class is "Bare soil."

**Algorithm 1** SASS Algorithm

**Input:** Input image $X$, manipulated label $\tilde{y}$, classifier $f$, loss function $L_{SASS}$, number of attack iterations $I$, perturbation magnitude $\epsilon$, patch size $s \times s$, step size $\eta$

**Output:** Perturbation vector $\delta$

$X_p \sim X$: $X_p \in \mathbb{R}^{s \times s \times c}$

**for all** $X_p$ **do**

  $\delta_0 \leftarrow 0$

  **for** $i = 0$ to $I - 1$ **do**

    $\delta_{i+1} \leftarrow \delta_i - \eta \cdot \nabla_{\delta_i} L_{SASS}\left(X_p + \delta_i, \tilde{y}\right)$

    $\delta_{i+1} \leftarrow Clamp\left(\epsilon \cdot \frac{\delta_{i+1}}{\|\delta_{i+1}\|_\infty}\right)$

  **end for**

**end for**

**return** $\delta = \delta_I$

where $\alpha$ is a hyperparameter that controls the tradeoff between the two loss components. With the loss function defined above, we can optimize the perturbation $\delta$. The detailed of SASS is summarized in Algorithm 1.

### C. Masked Stealthy Attack for Semantic Segmentation

While showing strong attack abilities, SASS can only produce adversarial perturbations on the exact input patch when the segmentation models are accessible. In practice, however, different mechanisms to crop the high-resolution remote sensing images can be applied, e.g., overlapped inference. As shown in Fig. 3, we choose the trihedral building as the source class and "Bare soil" as the target class. SASS successfully fools the model to predict the "trihedral building" as "Bare soil." Nevertheless, when overlapped inference is applied, the structures of adversarial perturbations generated by SASS are broken, leading to the failure of SASS. To make the adversarial examples resistant to overlapped inferences or other potential transformations, we further propose the MSASS that focuses on perturbing the exact victim objects only.

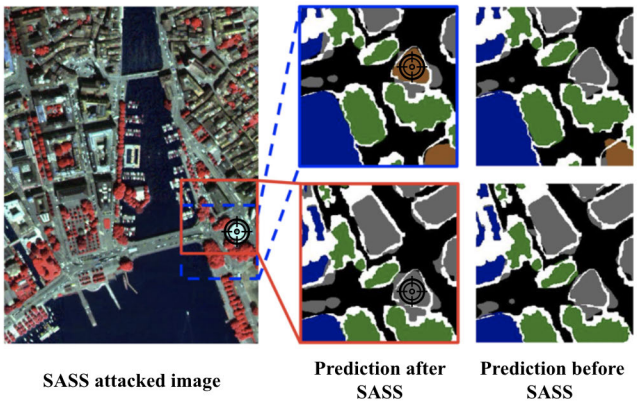



Fig. 3. Segmentation results of SASS on the Zurich Summer dataset using the ICNet model during overlapped inference. The figure shows the segmentation results of the attacked patch during training (top left), in which the trihedral building is misclassified as "Bare soil," and the segmentation results of a different interested patch with an overlapping area to the trained one (bottom left), in which the attack on the trihedral building is unsuccessful.

The goal of MSASS is to make the segmentation models misclassify the victim objects in any patch where the victim areas exist. Based on SASS, we first produce a set of patches for generating perturbations and then employ a mask to make sure that perturbations are only applied to the source victim area. The masked perturbation, denoted as $\delta_{\text{masked}}$, is obtained by element-wise multiplication between the collective perturbation vector $\delta$ and binary mask $M$, as shown in the following equation:

$$\delta_{\text{masked}} = \delta \odot M \tag{6}$$

where $\odot$ denotes the element-wise multiplication and $M_i = 1$ if pixel $i$ belongs to the victim area and $M_i = 0$ otherwise. Instead of finding the patches containing the victim source object iteratively by examining each pixel, a more efficient modification is made in that we utilize the minimum overlap used to generate a grid-like patch layout, as illustrated in Fig. 4. This modification (e.g., setting the minimum overlap to 200) ensures that patches cover each direction and

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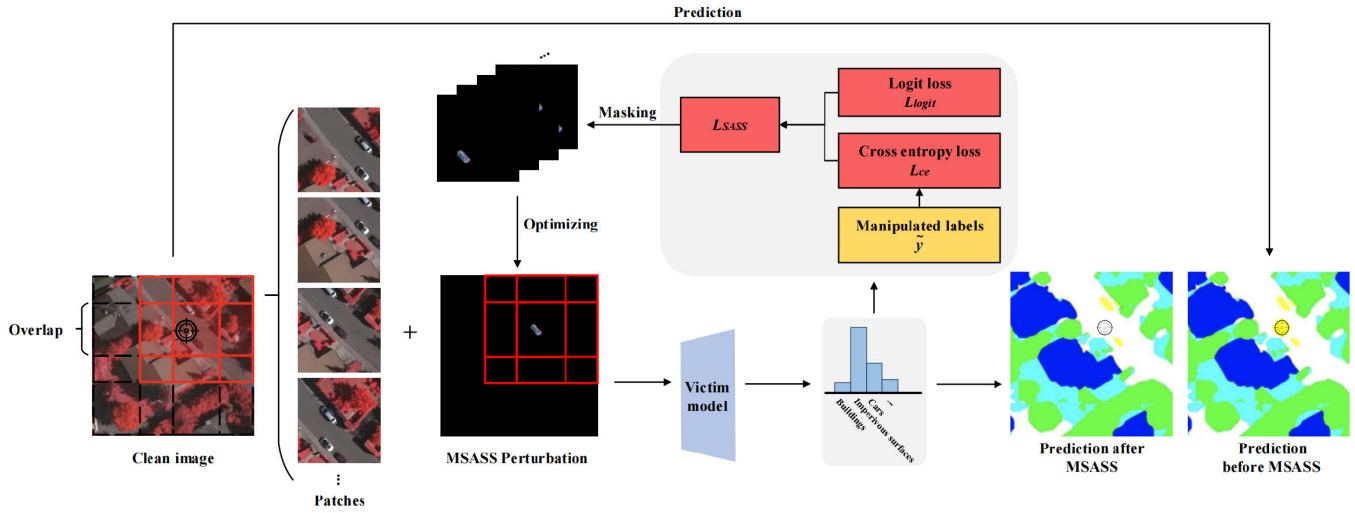



Fig. 4. Overview of our proposed MSASS model. The victim instance is the aimed white car with a black roof. The target class is "impervious surface." The red border marks the selected patch for constructing the adversarial example. The collective perturbation is added to each selected patch to produce the adversarial example.

position containing the victim source areas. Specifically, in our approach, an efficient and distinctive feature is the absence of an additional patch division process beyond the initial one. During the initial patching process, a straightforward strategy involves setting the minimum overlap to a higher threshold, such as 200. This modification yields multiple patches encompassing the targeted object. In the subsequent selection step, all patches containing the entirety of the targeted object are chosen, as exemplified in Fig. 4. We also employ the $L_{\text{SASS}}$ loss function to optimize the collective perturbation vector in the masked stealthy attack. The final objective function of MSASS can be formulated as follows:

$$\arg\min_{\delta} \frac{1}{m}\sum_{i=0}^{m}(1-\alpha)L_{\text{logit}}(\mathcal{S}_i(X_{\text{adv}})) + \alpha L_{\text{ce}}\left(\mathcal{S}_i(X_{\text{adv}}), \tilde{y}\right)$$
$$X_{\text{adv}} = X + \delta_{\text{masked}} \tag{7}$$

where $\mathcal{S}(\cdot)$ denotes the set of image patches in which the source victim area exists and $\mathcal{S}_i(\cdot)$ represents the $i$th image patch within the set. The total number of patches used for generating perturbations is denoted as $m$. The details of MSASS can be found in Algorithm 2.

**Algorithm 2** MSASS Algorithm

**Input:** Input image $X$, manipulated label $\tilde{y}$, classifier $f$, loss function $L_{SASS}$, number of attack iterations $I$, perturbation magnitude $\epsilon$, patch size $s \times s$, step size $\eta$
**Output:** Masked perturbation vector $\delta_{masked}$
$X_p \sim X$: $X_p \in \mathbb{R}^{s\times s\times c}$
$\delta_0 \leftarrow 0$
**for** $i = 0$ to $I-1$ **do**
  $L_{total} \leftarrow 0$
  **for all** $X_p$ **do**
    **if** victim source object in $X_p$ **then**
      $L_{total} += L_{SASS}(X_p + \delta_i, \tilde{y})$
    **end if**
  **end for**
  $\delta_{i+1} \leftarrow \delta_i - \eta \cdot \nabla_{\delta_i} L_{total}$
  $\delta_{i+1} \leftarrow Clamp\left(\epsilon \cdot \frac{\delta_{i+1}}{\|\delta_{i+1}\|_\infty}\right)$
  $\delta_{i+1} \leftarrow Mask(\delta_{i+1})$
**end for**
**return** $\delta_{masked} = \delta_I$

## IV. Experiments

*1) Datasets:* We conducted experiments using two remote sensing image datasets that are publicly available: Vaihingen [57] and Zurich Summer [58].

The Vaihingen dataset, which is a subset of the data used by the German Society of Photogrammetry, Remote Sensing and Geoinformation (DGPF) to test digital aerial cameras, is a well-known benchmark dataset for semantic segmentation in remote sensing [57]. The spatial resolution is 9 cm and the average size of images is around 2500 × 1900 pixels. Out of a total of 33 images of Vaihingen city, 16 are meticulously annotated, delineating six distinct land-cover categories: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. To ensure comparability with prior studies [12], we adopt the same train–test partitioning strategy, where five images (image IDs: 11, 15, 28, 30, and 34) are reserved for testing, while the remaining images constitute the training set.

The Zurich Summer dataset consists of 20 images, which were captured over Zurich city in August 2002 [58]. The spatial resolution is 0.62 m, and the average size of an image is around 1000 × 1000 pixels. The dataset is comprised of eight distinct urban classes, including roads, buildings, trees, grass, bare soil, water, rails, and pools. Pixels that are not classified as any of the eight urban classes are labeled as background. In accordance with prior research [12], we select five images (image IDs: 16, 17, 18, 19, and 20) as the test set, while the other 15 images are used as the training set.

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TABLE I

EXPERIMENTAL RESULTS FOR SASS ON THE VAIHINGEN DATASET. FOR EACH ATTACK SCENARIO, THE SOURCE-TO-TARGET RATIO ($R_s$) AND OTHERS-TO-TARGET RATIO ($R_{\text{NS}}$) ARE REPORTED. THE HIGHER (LOWER) $R_s$ ($R_{\text{NS}}$), THE BETTER. ALL NUMBERS ARE REPORTED IN %

| Attack Scenario | FCN8s | | DeepLabV3+ | | U-Net | | ICNet | | Unetformer | | ST-Unet | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ |
| Cars → Impervious surfaces | 99.66 | 0.06 | 99.83 | 0.05 | 99.41 | 0.05 | 99.81 | 0.03 | 99.07 | 0.05 | 99.94 | 0.04 |
| Impervious surfaces → Cars | 99.66 | 0.24 | 99.87 | 0.14 | 99.80 | 0.12 | 99.97 | 0.23 | 99.69 | 0.10 | 99.77 | 0.06 |
| Cars → Buildings | 98.86 | 0.24 | 99.40 | 0.14 | 99.87 | 0.17 | 99.82 | 0.12 | 99.98 | 0.43 | 99.94 | 0.20 |
| Buildings → Low vegetation | 99.93 | 0.68 | 99.97 | 0.48 | 99.87 | 0.44 | 99.98 | 0.34 | 99.85 | 0.22 | 99.83 | 0.12 |
| Low vegetation → Buildings | 99.09 | 0.46 | 99.73 | 0.17 | 99.90 | 0.18 | 99.84 | 0.20 | 97.24 | 0.15 | 99.26 | 0.06 |

TABLE II

EXPERIMENTAL RESULTS FOR SASS ON THE ZURICH SUMMER DATASET. FOR EACH ATTACK SCENARIO, THE SOURCE-TO-TARGET RATIO ($R_s$) AND OTHERS-TO-TARGET RATIO ($R_{\text{NS}}$) ARE REPORTED. THE HIGHER (LOWER) $R_s$ ($R_{\text{NS}}$), THE BETTER. ALL NUMBERS ARE REPORTED IN %

| Attack Scenario | FCN8s | | DeepLabV3+ | | U-Net | | ICNet | | Unetformer | | ST-Unet | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ |
| Buildings → Bare soil | 72.80 | 0.28 | 85.40 | 0.05 | 77.13 | 0.03 | 99.53 | 0.22 | 97.78 | 0.01 | 99.07 | 0.02 |
| Bare soil → Buildings | 96.32 | 0.02 | 99.78 | 0.00 | 99.89 | 0.01 | 99.94 | 0.00 | 81.40 | 0.09 | 93.86 | 0.04 |
| Buildings → Roads | 99.13 | 0.06 | 99.75 | 0.02 | 99.31 | 0.04 | 99.92 | 0.06 | 96.30 | 0.10 | 99.57 | 0.09 |
| Roads → Buildings | 99.32 | 0.23 | 99.94 | 0.10 | 99.55 | 0.05 | 99.95 | 0.09 | 95.54 | 0.13 | 99.18 | 0.20 |
| Buildings → Water | 80.59 | 0.10 | 99.82 | 0.02 | 99.95 | 0.04 | 99.28 | 0.04 | 99.68 | 0.03 | 99.81 | 0.06 |

*2) Training Details:* For the following experiments, we fix the number of attack iterations to $I = 500$ and adopt the $l_\infty$ norm for the calculation of perturbation. We cap the maximized perturbation at $\epsilon = 15$ for images in the range [0, 255]. In addition, we constrain the magnitude of the manipulated image to lie in the permissible image range [0, 1]. The attack step size is 0.01. Unless otherwise mentioned, we experiment with $d = 10$ and $\alpha = 0.1$ for the calculation of $L_{\text{SASS}}$. In our experiments, we employ a patching process of the high-resolution images, using a fixed patch size of $256 \times 256$ pixels. In the training of the SASS, we set the minimum overlap of the patching process to 0 pixels. For the training of the MSASS, we increase the minimum overlap of the patching process to 200 pixels. This ensures that the patches used during training have a substantial overlap with surrounding patches for collective training. We conduct all the experiments with PyTorch on a single GPU NVIDIA GeForce RTX 2080 Ti.

*3) Metrics:* In order to assess the effectiveness of our method, we utilize the metric of the source-to-target ratio, denoted as $R_s$, which is determined by calculating the proportion of samples from the source class being successfully misclassified into the target class, and the others-to-target ratio, denoted as $R_{\text{ns}}$, which is determined by calculating the proportion of samples that are not from the source class fooled into the target class. Consequently, the higher (lower) $R_s$ ($R_{\text{ns}}$), the better. The overall accuracy $\text{OA} = n_{\text{correct}}/n_{\text{total}}$ is adopted to measure the segmentation performance of different models, where $n_{\text{correct}}$ denotes the number of correctly classified pixels.

### A. SASS Attack Performance

*1) Quantitative Results:* We assess the proposed SASS with four state-of-the-art CNN-based semantic segmentation networks: FCN8s [43], U-Net [44], ICNet [47], DeepLabV3+ [45], and two transformer-based networks, which are Unetformer [22] and ST-Unet [33]. We randomly select five source-to-target attack scenarios for each dataset and denote each scenario by "Source class → Target class."

TABLE III

OVERALL ACCURACY COMPARISON FOR THE SASS AND CLEAN PREDICTIONS ON THE VAIHINGEN DATASET. THE RESULTS ARE REPORTED IN OVERALL ACCURACY %. I: IMPERVIOUS SURFACES. C: CARS

| Method | w/o attack | I → C | C → I |
|---|---|---|---|
| FCN8s | 84.60 | 83.20 | 84.05 |
| U-Net | 84.17 | 84.52 | 83.15 |
| DeepLabV3+ | 83.75 | 83.07 | 83.98 |
| ICNet | 83.73 | 82.84 | 84.09 |
| Unetformer | 85.60 | 84.90 | 85.75 |
| ST-Unet | 84.51 | 83.77 | 84.81 |

The SASS attack results on two datasets are summarized in Tables I and II, where we report the source-to-target ratio ($R_s$) and others-to-target ratio ($R_{\text{ns}}$) for each attack scenario.

In general, we can observe that $R_s$ is notably high (even close to 100%) for most scenarios on both CNN- and transformer-based semantic segmentation networks. For instance, the $R_s$ value for the "Cars → Buildings" scenario in the Unetformer network attains 99.98%, accompanied by $R_{\text{ns}}$ of only 0.43%. This indicates that SASS can manipulate the pixels from the source class to be predicted as the target class with a high success rate. Furthermore, $R_{\text{ns}}$ is low (below 1%) for all scenarios on both datasets and across both CNN- and transformer-based semantic segmentation networks, which demonstrates that the SASS can distinguish the victim source pixels from the others and minimize the attack impact on nonsource pixels.

To evaluate the effectiveness of SASS in mitigating the adversarial impact on nonsource pixels, we compare the overall accuracy of SASS-attacked models with models without any attacks. For each dataset, we select two attack scenarios

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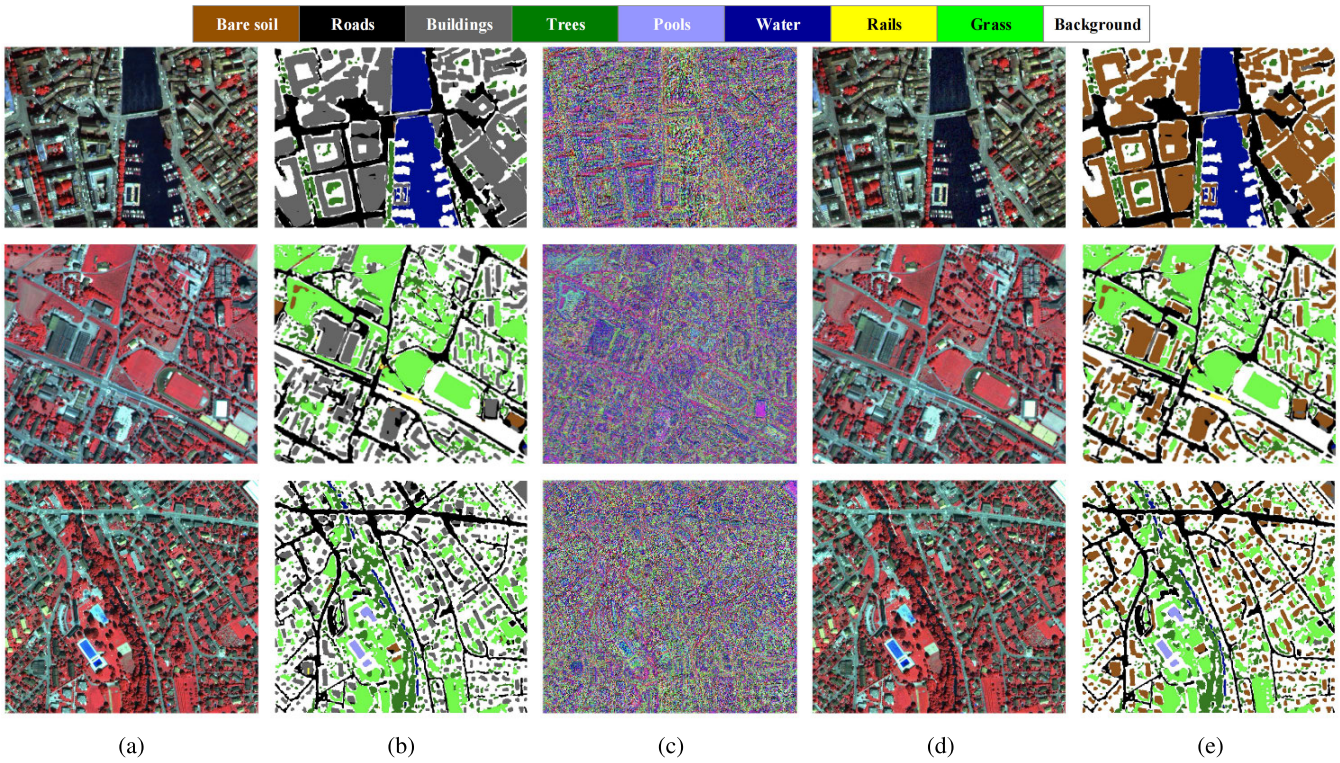



Fig. 5. Qualitative results of the SASS in the Zurich Summer dataset on ICNet. (a) Original clean test images in the Zurich Summer dataset. (b) Segmentation results of ICNet on the clean images. (c) "Buildings → Bare soil" SASS perturbations. (d) Corresponding adversarial examples. (e) Segmentation results on the adversarial images.

TABLE IV
OVERALL ACCURACY COMPARISON FOR THE SASS AND CLEAN PREDICTIONS ON THE ZURICH SUMMER DATASET. THE RESULTS ARE REPORTED IN OVERALL ACCURACY %. R: ROADS. B: BUILDINGS

| Method | w/o attack | R → B | B → R |
|---|---|---|---|
| FCN8s | 90.11 | 89.59 | 89.82 |
| U-Net | 91.94 | 93.35 | 90.92 |
| DeepLabV3+ | 92.48 | 91.34 | 93.20 |
| ICNet | 91.84 | 90.09 | 91.75 |
| Unetformer | 92.56 | 92.24 | 92.16 |
| ST-Unet | 91.27 | 91.50 | 90.79 |

and evaluate the overall accuracy of nonsource pixels in the SASS-attacked predictions. We also assess the overall accuracy of models without any attacks for comparison. The comparison results are summarized in Tables III and IV. The results reveal that the overall accuracy drop after applying SASS is less than 2%. Notably, in certain cases such as the "Buildings → Roads" SASS attack on the DeepLabV3+ model, the overall accuracy of nonsource pixels even surpasses that of the overall clean predictions. The observed phenomenon of higher overall accuracy performance on the nonsource pixels within attacked images compared to the original clean images can be elucidated through a dual perspective. First, while the adversarial perturbation trained on the nonsource pixels is designed to preserve the initial prediction, this preservation goal is also subject to the influence of adjacent source pixel predictions. The choice of the sink class as logically inconspicuous enhances the likelihood of correct predictions for nonsource pixels due to the model's inherent characteristics. Second, this discrepancy may arise from a potentially difficult prediction of the source pixels in the original clean image. These findings suggest that our SASS can effectively mitigate the impact of attacks on nonsource pixels.

*2) Qualitative Results:* We show the qualitative results of the SASS in the Zurich Summer dataset on ICNet in Fig. 5, including the original clean test images, the segmentation results of ICNet on the clean images, the adversarial perturbation, the corresponding adversarial examples, and the segmentation results of ICNet on the adversarial images. The attack scenario considered is "Buildings → Bare soil." We can observe that the victim source pixels have been predicted as the "Bare soil" instead of "Buildings." At the same time, the segmentation results of those nonsource context pixels are preserved. These findings provide evidence of the effectiveness of the SASS in inducing stealthy but effective changes to the segmentation output.

There exists a body of advanced works [55], [56], [59] that have proposed targeted attack losses. Given that one of our key contributions is the design of the SASS loss, it is imperative to benchmark and compare it against other targeted attack losses. Although these losses are originally developed for image classification tasks rather than semantic segmentation, we posit that they can be adapted for this purpose with minimal modifications. Specifically, each pixel in semantic segmentation is treated as an image in image classification. For our comparative analysis, we selected the following advanced targeted attack losses: CE loss, Po+Trip loss [59], and logit loss [55]. We used the default hyperparameters for

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TABLE V

ATTACK PERFORMANCE COMPARISON OF DIFFERENT LOSS FUNCTIONS ON ZURICH SUMMER AND VAIHINGEN DATASETS. FOR EACH ATTACK SCENARIO, THE SOURCE-TO-TARGET RATIO ($R_s$) AND OTHERS-TO-TARGET RATIO ($R_{\text{NS}}$) ARE REPORTED. THE HIGHER (LOWER) $R_s$ ($R_{\text{NS}}$), THE BETTER. ALL NUMBERS ARE REPORTED IN %

| Dataset | Attack Scenario | Metric | Unetformer | | | | U-Net | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | CE | Logit | Po+Trip | SASS | CE | Logit | Po+Trip | SASS |
| Zurich Summer | Bare soil → Building | $R_s(\uparrow)$ | 91.41 | 100.00 | 89.78 | 98.17 | 97.67 | 100.00 | 96.99 | 99.90 |
| | | $R_{ns}(\downarrow)$ | 0.00 | 3.10 | 0.02 | 0.01 | 0.01 | 1.50 | 0.02 | 0.01 |
| | Building → Bare soil | $R_s(\uparrow)$ | 33.14 | 82.87 | 8.51 | 76.15 | 61.65 | 83.58 | 18.29 | 77.38 |
| | | $R_{ns}(\downarrow)$ | 0.01 | 2.73 | 0.02 | 0.02 | 0.02 | 1.09 | 0.01 | 0.09 |
| Vaihingen | Cars → Impervious surfaces | $R_s(\uparrow)$ | 92.87 | 100.00 | 97.28 | 98.70 | 97.65 | 99.74 | 88.03 | 99.41 |
| | | $R_{ns}(\downarrow)$ | 0.05 | 4.98 | 0.13 | 0.06 | 0.05 | 0.68 | 0.26 | 0.05 |
| | Impervious surfaces → Cars | $R_s(\uparrow)$ | 98.34 | 99.98 | 96.04 | 99.60 | 98.48 | 99.96 | 88.07 | 99.76 |
| | | $R_{ns}(\downarrow)$ | 0.02 | 13.49 | 0.08 | 0.10 | 0.01 | 10.12 | 0.06 | 0.13 |

our competitors. For the Po+Trip loss, the hyperparameters are set as follows: $\xi = 1e^{-5}$, $\lambda = 1e^{-2}$, and $\gamma = 7 \times 1e^{-3}$. Moreover, the attack frameworks employed in our experiments leverage the PGD method [9]. Consistency is maintained by setting the other attack hyperparameters to the same values as those proposed for SASS. The victim models for our experiments are U-Net and Unetformer. We conducted experiments on both the Vaihingen dataset and the Zurich Summer dataset. Each dataset is subjected to two source-to-target attack scenarios, denoted as "Source class → Target class." The comprehensive results from these experiments are summarized in Table V. The comparative analysis of different loss functions reveals distinct tradeoffs in their impact on targeted adversarial attacks. Logit loss emerges with the highest $R_s$, underscoring its profound effect in transforming source pixels into the target class. However, this advantage is tempered by logit loss having the largest $R_{\text{ns}}$ among the compared losses, indicating a deficiency in preserving predictions for nonsource pixels and resulting in a less stealthy attack. Contrastingly, CE loss exhibits a weaker $R_s$ compared to both our proposed SASS loss and logit loss. Despite this, CE loss outperforms logit loss in $R_{\text{ns}}$, signifying its ability to achieve a more stealthy attack by better-preserving predictions for nonsource pixels. Despite having a less dominant source-to-target ratio, CE loss excels in maintaining the integrity of nonsource predictions. The Po+Trip loss achieves a certain level of performance in our stealthy attack scenario, achieving a more stealthy attack than Logit loss, albeit with a weaker source-to-target ratio. Our proposed SASS loss demonstrates comprehensive superiority in both $R_s$ and $R_{\text{ns}}$. By integrating principles from logit loss and CE loss, our proposed SASS loss strikes a balance between maximizing attack success and maintaining a stealthy effect. The comparative analysis is consistent with the findings of our final ablation study, which establishes that logit loss has the most significant impact on attacking source pixels, while the CE loss plays a critical role in preserving nonsource pixel predictions to achieve a more stealthy attack.

### B. MSASS Attack Performance

*1) Quantitative Results:* To evaluate the effectiveness of our MSASS approach under overlapped inference, we partition the high-resolution remote sensing images into overlapping patches that contain the victim area using different test minimum overlaps (0, 100, and 200). In our experiments, we utilize four state-of-the-art CNN-based semantic segmentation networks: FCN8s [43], U-Net [44], ICNet [47], and DeepLabV3+ [45]. In addition, we incorporate two state-of-the-art transformer-based networks, which are Unetformer [22] and ST-Unet [33], as our pretrained victim models.

For each dataset, we randomly select two source-to-target attack scenarios, denoted as "Source class → Target class," and choose one victim source object for each test image. To ensure fairness in the comparison, the same victim source object is used for each pretrained model. The results obtained from these experiments are summarized in Table VI, where we present the source-to-target ratio ($R_s$) and others-to-target ratio ($R_{\text{ns}}$) for each attack scenario across all patches that contain the victim area.

Our MSASS demonstrates effective attack performance when applied in overlapped inference with an average $R_s$ exceeding 84%, indicating successful manipulation of the victim source object to the target class. The average $R_{\text{ns}}$ remains below 1%, indicating minimal impact on nonvictim areas during overlapped inference. However, it should be noted that the efficacy of MSASS on the ICNet model is relatively lower. This reduced performance can be attributed to the cascade image inputs employed by ICNet, where the downsampling process may interfere with the perturbation introduced by MSASS, thereby diminishing its overall effectiveness.

In addition, we compare the attack performance of our MSASS and SASS methods under overlapped inference, as shown in Table VII. To perform the comparison, we translate the exact input patch by 50 pixels both horizontally and vertically to generate the overlapped patch for evaluation. The results clearly demonstrate that MSASS consistently outperforms SASS, exhibiting significantly higher $R_s$ values. This indicates the superior ability of MSASS to launch a more robust attack that is less susceptible to variations in the surrounding context of victim objects during overlapped inference. For instance, in the "Buildings → Low vegetation" attack scenario, when overlapped inference is applied, SASS achieves $R_s$ of only 68.15%, while MSASS achieves an impressive $R_s$ of 95.01%.

*2) Qualitative Results:* Fig. 6 presents the qualitative results of the MSASS method on the FCN8s model using the

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TABLE VI

ATTACK PERFORMANCE OF MSASS WITH DIFFERENT TEST MINIMUM OVERLAP THRESHOLDS IN THE VAIHINGEN AND ZURICH SUMMER DATASETS. FOR EACH ATTACK SCENARIO, THE SOURCE-TO-TARGET RATIO ($R_s$) AND OTHERS-TO-TARGET RATIO ($R_{\text{NS}}$) ARE REPORTED. THE HIGHER (LOWER) $R_s$ ($R_{\text{NS}}$), THE BETTER. ALL NUMBERS ARE REPORTED IN %

| Dataset | Attack Scenario | Model | Min Overlap = 0 | | Min Overlap = 100 | | Min Overlap = 200 | |
|---|---|---|---|---|---|---|---|---|
| | | | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ |
| Vaihingen | Cars → Impervious surfaces | FCN8s | 93.62 | 0.30 | 92.48 | 1.00 | 96.78 | 0.99 |
| | | U-Net | 91.88 | 0.32 | 88.73 | 0.76 | 93.36 | 1.74 |
| | | DeepLabV3+ | 71.36 | 0.29 | 74.27 | 1.88 | 80.95 | 2.26 |
| | | ICNet | 62.56 | 0.43 | 65.00 | 1.75 | 80.92 | 2.27 |
| | | Unetformer | 80.35 | 0.26 | 77.81 | 1.15 | 80.75 | 1.26 |
| | | ST-Unet | 74.34 | 0.94 | 74.33 | 1.07 | 92.30 | 1.49 |
| | Buildings → Low vegetation | FCN8s | 98.23 | 1.00 | 97.28 | 1.43 | 98.77 | 1.60 |
| | | U-Net | 79.28 | 0.24 | 76.52 | 1.23 | 91.13 | 1.11 |
| | | DeepLabV3+ | 80.05 | 0.71 | 64.89 | 1.73 | 93.99 | 1.48 |
| | | ICNet | 78.92 | 0.46 | 62.55 | 0.76 | 92.93 | 1.32 |
| | | Unetformer | 94.75 | 1.09 | 94.58 | 1.34 | 97.93 | 1.76 |
| | | ST-Unet | 74.33 | 0.80 | 75.61 | 3.22 | 97.53 | 2.18 |
| Zurich Summer | Buildings → Roads | FCN8s | 90.91 | 0.07 | 85.98 | 0.64 | 89.55 | 0.83 |
| | | U-Net | 96.07 | 0.13 | 89.29 | 0.56 | 95.38 | 0.35 |
| | | DeepLabV3+ | 96.21 | 0.01 | 92.46 | 0.37 | 97.44 | 0.38 |
| | | ICNet | 79.67 | 0.17 | 58.56 | 1.00 | 85.99 | 0.98 |
| | | Unetformer | 88.46 | 0.45 | 70.20 | 1.57 | 84.86 | 0.63 |
| | | ST-Unet | 84.14 | 0.63 | 74.13 | 1.16 | 92.67 | 1.81 |
| | Bare soil → Buildings | FCN8s | 77.40 | 0.27 | 82.34 | 0.61 | 76.57 | 0.27 |
| | | U-Net | 93.74 | 0.15 | 81.21 | 1.54 | 92.87 | 1.74 |
| | | DeepLabV3+ | 92.94 | 1.62 | 86.04 | 1.90 | 86.36 | 1.16 |
| | | ICNet | 82.15 | 1.04 | 71.85 | 1.69 | 70.78 | 2.08 |
| | | Unetformer | 99.47 | 1.33 | 91.55 | 2.01 | 96.70 | 3.32 |
| | | ST-Unet | 96.85 | 0.96 | 81.71 | 1.13 | 91.74 | 0.67 |

Vaihingen dataset. The chosen victim source object is a black car, and the attack scenario considered is "Car → Impervious surfaces." In this figure, we display three overlapping patches of the black car, along with the corresponding segmentation results before and after the MSASS attack. In addition, we show the perturbation applied and the resulting adversarial images. It is important to note that the perturbations applied to the same victim source object through different overlapping patches are the same. The observation from the prediction results is that the victim source object, the black car, is misclassified as "Impervious surfaces." Conversely, the other white car in the image, which is not the targeted object, maintains its original semantic label as "Car." This finding emphasizes the effectiveness of the MSASS attack in selectively altering the predictions for the victim source object while minimizing the impact on other objects in the scene.

### C. Evading Spatial Consistency Detection

Spatial consistency analysis, introduced by Xiao et al. [1], is known as a technique for detecting adversarial examples in the domain of semantic segmentation. In this section, we evaluate the efficacy of our proposed SASS and MSASS in evading detection based on spatial consistency analysis.

For evaluation, we conducted experiments on the Zurich dataset using four pretrained state-of-the-art CNN-based semantic segmentation models and two transformer-based networks, which are Unetformer [22] and ST-Unet [33], considering all five attack scenarios mentioned in Table II. Following previous settings, we set the number of overlapped regions to 50, with the lower bound of the region defined as $b_{\text{low}} = 16$ and the upper bound as $b_{\text{upper}} = 32$. To determine the threshold for the detection method, we recorded the lowest bound of the mean intersection over union (mIOU) [1] for each test image. It is worth noting that this threshold is more stringent compared to the threshold used in the original work, which was learned from a set of benign training data. Given an image, we calculate the mIOU for each overlapping region of two random patches and compare it with the previously calculated threshold. If the mIOU is larger than the threshold, the image is recognized as benign; otherwise, it is considered an adversarial example. To ensure an equal number of test samples, we performed segmentation on the input images for the targeted PGD500 (T-PGD500) attack method and untargeted PGD500 (UT-PGD500) attack method.

The results of our evaluation, including the detection success rates of adversarial examples for each pretrained victim model and the total detection success rates of adversarial examples across all victim models, are presented in Table VIII. The experimental results reveal that the spatial detection method achieves a notable detection success rate of up to 97.78% for untargeted PGD500 attacks. When subjected to our proposed SASS, the detection success rate drops to 54.65%, demonstrating its ability to partially deceive the spatial consistency analysis. Notably, SASS on Unetformer demonstrates the most stealthy performance with a detection success rate of only 33.34%. Furthermore, the proposed MSASS exhibits even stronger evasion capabilities generally, with a significantly reduced detection success rate of only 28.71%. Among all the six semantic segmentation networks, MSASS on ICNet

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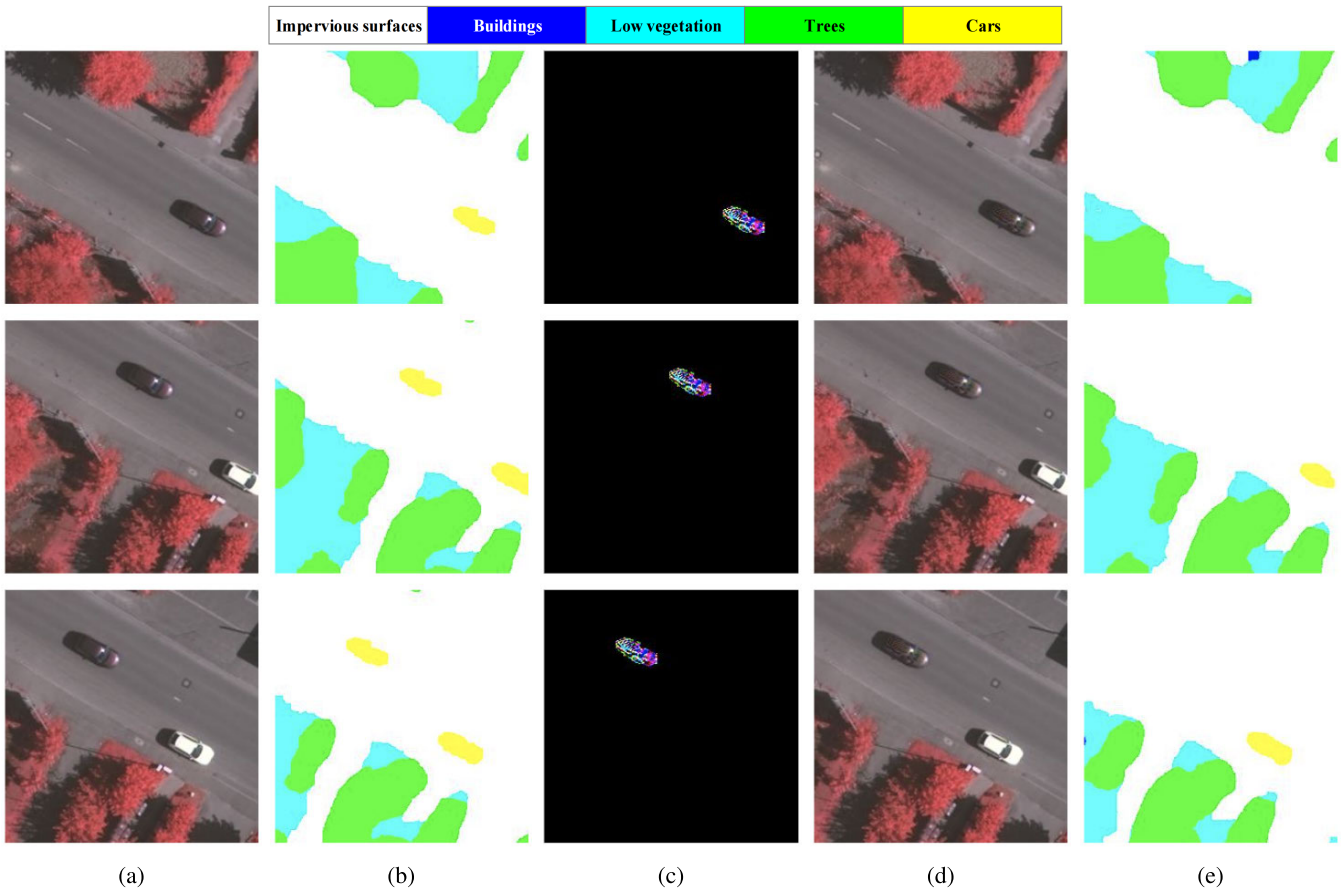



Fig. 6. Qualitative results of the MSASS in the Vaihingen dataset on FCN8s. (a) Original clean test images in the Vaihingen dataset. (b) Segmentation results of FCN8s on the clean images. (c) "Car → Impervious surfaces" MSASS perturbations. (d) Corresponding adversarial examples. (e) Segmentation results of the adversarial images.

demonstrates the most stealthy performance, with a detection success rate of only 18.18%. These findings highlight the effectiveness of our proposed methods in bypassing spatial consistency analysis and underscore the need for enhanced defense mechanisms to mitigate the impact of such adversarial attacks in semantic segmentation tasks.

### D. Ablation Analysis

We conducted an ablation analysis to investigate the impact of different design choices of the proposed stealthy attack loss function, denoted as $L_{\text{SASS}}$, using the ICNet model in the Zurich Summer dataset, which is summarized in Table IX. The chosen attack scenario is "Buildings → Bare soil." The remaining experimental settings remain consistent with those employed previously.

It can be observed that the $L_{\text{SASS}}$ loss function exhibits the most favorable performance, as indicated by the largest gap of 98.61% between $R_s$ and $R_{\text{ns}}$ metrics. This outcome underscores the effectiveness of $L_{\text{SASS}}$ in selectively manipulating the victim source class to the target class while minimizing the impact on nonvictim pixels. When excluding the $L_{\text{logit}}$ component, the resulting $R_s$ values decrease from 98.78% to 83.40%, suggesting that $L_{\text{logit}}$ plays a significant role in shifting the source class to the target class. $L_{\text{ce}}$ represents the CE loss between the perturbed predictions and the target labels of all pixels. In constructing the target labels, if a pixel belongs to the source class, the manipulated label $\tilde{y}$ is flipped to the target class $y_t$, while it remains unchanged for nonsource pixels. Notably, when we remove the label flip of source pixels in $L_{\text{ce}}$, the resulting $R_s$ values decrease, indicating that this component contributes significantly to shifting the source class to the target class. Conversely, when we omit the nonsource pixels in $L_{\text{ce}}$, though achieve the highest $R_s$ of 99.63%, the $R_{\text{ns}}$ values increase more substantially, from 0.17% to 18.04%, suggesting that the inclusion of the $L_{\text{ce}}^{\text{nonsource}}$ term is crucial for preserving the predictions of nonsource pixels and achieving a stealthy attack.

### E. Hyperparameter Analysis

We performed a hyperparameter analysis to investigate the impact of the weighting factor $\alpha$ and dominance value $d$ on $R_s$ and $R_{\text{ns}}$ in SASS using the ICNet model in the Zurich Summer dataset. The chosen attack scenario is "Buildings → Roads." Moreover, we evaluate the impact of the minimum overlap parameter on $R_s$ and $R_{\text{ns}}$ in MSASS using the U-net and Unetformer model in the Vaihingen Summer dataset. The remaining experimental settings remain consistent with those employed previously.

*1) Weighting Factor $\alpha$:* In Fig. 7, we evaluate the impact of the weighting factor $\alpha$ on $R_s$ and $R_{\text{ns}}$. As $\alpha$ values increase,

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TABLE VII

ATTACK PERFORMANCE COMPARISON OF SASS AND MSASS UNDER OVERLAPPED INFERENCE. C: CARS, I: IMPERVIOUS SURFACES, V: LOW VEGETATION, R: ROADS, AND S: BARE SOIL. FOR EACH ATTACK SCENARIO, THE SOURCE-TO-TARGET RATIO ($R_s$) AND OTHERS-TO-TARGET RATIO ($R_{\text{NS}}$) ARE REPORTED. THE HIGHER (LOWER) $R_s$ ($R_{\text{NS}}$), THE BETTER. ALL NUMBERS ARE REPORTED IN %

| Dataset | Attack Scenario | Model | SASS | | MSASS | |
|---|---|---|---|---|---|---|
| | | | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ |
| Vaihingen | C → I | FCN8s | 72.39 | 0.56 | 91.94 | 1.17 |
| | | U-Net | 70.27 | 4.16 | 91.54 | 1.36 |
| | | DeepLabV3+ | 56.24 | 1.63 | 70.43 | 2.03 |
| | | ICNet | 53.98 | 1.31 | 81.10 | 2.39 |
| | | Unetformer | 62.31 | 0.31 | 71.26 | 0.96 |
| | | ST-Unet | 48.84 | 0.45 | 59.20 | 1.06 |
| | B → V | FCN8s | 68.15 | 1.71 | 95.01 | 1.06 |
| | | U-Net | 51.06 | 6.53 | 89.26 | 1.03 |
| | | DeepLabV3+ | 21.59 | 3.99 | 88.25 | 1.72 |
| | | ICNet | 19.24 | 2.59 | 54.65 | 1.19 |
| | | Unetformer | 79.00 | 0.77 | 81.40 | 1.36 |
| | | ST-Unet | 9.56 | 0.98 | 64.36 | 2.51 |
| Zurich Summer | B → R | FCN8s | 65.04 | 0.36 | 89.57 | 0.97 |
| | | U-Net | 15.15 | 0.71 | 85.57 | 0.47 |
| | | DeepLabV3+ | 19.55 | 0.69 | 89.64 | 0.69 |
| | | ICNet | 6.93 | 1.06 | 65.75 | 1.45 |
| | | Unetformer | 40.75 | 0.24 | 53.07 | 1.05 |
| | | ST-Unet | 16.20 | 0.13 | 75.73 | 1.99 |
| | S → B | FCN8s | 54.18 | 0.22 | 79.21 | 0.43 |
| | | U-Net | 30.93 | 0.72 | 83.08 | 1.32 |
| | | DeepLabV3+ | 17.56 | 1.47 | 69.05 | 1.11 |
| | | ICNet | 2.33 | 1.02 | 47.24 | 1.55 |
| | | Unetformer | 48.48 | 0.27 | 55.56 | 4.35 |
| | | ST-Unet | 38.26 | 0.42 | 72.95 | 0.58 |

TABLE VIII

COMPARISON OF DETECTION SUCCESS RATES ON THE ZURICH SUMMER DATASET. ALL NUMBERS ARE REPORTED IN %

| | T-PGD500 | UT-PGD500 | SASS | MSASS |
|---|---|---|---|---|
| FCN8s | 35.00 | 100.00 | 40.00 | 22.73 |
| U-Net | 75.00 | 86.67 | 68.18 | 22.73 |
| DeepLabV3+ | 25.00 | 100.00 | 81.82 | 40.91 |
| ICNet | 100.00 | 100.00 | 63.64 | 18.18 |
| Unetformer | 100.00 | 100.00 | 33.34 | 45.00 |
| ST-Unet | 100.00 | 100.00 | 40.90 | 22.73 |
| Average | 72.50 | 97.78 | 54.65 | 28.71 |

TABLE IX

ABLATION ANALYSIS. ATTACK PERFORMANCE OF "BUILDINGS → BARE SOIL" IN THE ZURICH SUMMER DATASET ON THE ICNET MODEL. THE SOURCE-TO-TARGET RATIO ($R_s$) AND OTHERS-TO-TARGET RATIO ($R_{\text{NS}}$) ARE REPORTED. THE HIGHER (LOWER) $R_s$ ($R_{\text{NS}}$), THE BETTER. ALL NUMBERS ARE REPORTED IN %

| Loss | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ |
|---|---|---|
| $L_{SASS}$ | 98.78 | 0.17 |
| w/o $L_{ce}^{label-flip}$ | 82.20 | 0.13 |
| w/o $L_{ce}^{non-source}$ | 99.63 | 18.04 |
| w/o $L_{logit}$ | 83.40 | 0.02 |

$R_s$ decreases generally. This is mainly because $\alpha$ weighs $L_{\text{ce}}$, which also emphasizes the preservation of nonvictim pixel predictions. Furthermore, an increase in the value of $\alpha$ decreases the value of $R_{\text{ns}}$, indicating that fewer nonvictim pixels have been shifted to the target class. Notably, a sharp decrease in $R_{\text{ns}}$ is observed for $\alpha$ values between 0 and 0.0001, reaching a consistently low $R_{\text{ns}}$ thereafter. To maximize the success rate of source pixels being classified as the target class and minimize the impact on other pixels, achieving a large gap between $R_s$ and $R_{\text{ns}}$ is essential. Based on our findings, an $\alpha$ value around 0.1 may be more beneficial for an effective attack.

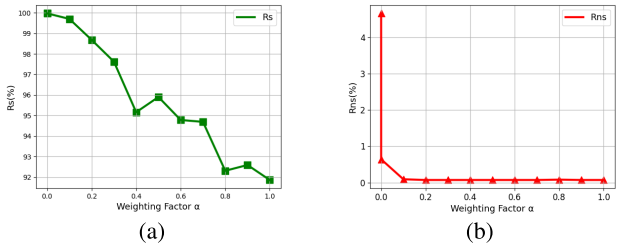



Fig. 7. Analysis of the influence of the weighting factor $\alpha$ on (a) $R_s$ and (b) $R_{\text{ns}}$ in the Zurich Summer dataset. The higher (lower) $R_s$ ($R_{\text{ns}}$), the better.

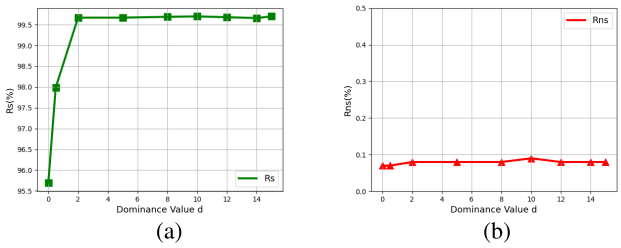



Fig. 8. Results of the impact of $d$ on (a) $R_s$ and (b) $R_{\text{ns}}$ in the Zurich Summer dataset. The higher (lower) $R_s$ ($R_{\text{ns}}$), the better.

*2) Dominance Value d:* In Fig. 8, we evaluate the impact of the dominance value $d$ on $R_s$ and $R_{\text{ns}}$. The dominance value $d$ is introduced to regulate the discrepancy between the predicted logit of the target class and the maximized logit of other classes in (3). Our experimental findings demonstrate a notable increase in $R_s$ as $d$ varies from 0 to 2, reaching a steadily high $R_s$ thereafter. Notably, when we set $d = 10$, the highest $R_s$ value is attained, indicating the successful shift of the majority of source pixel predictions to the target class. It is important to note that a higher value of $d$ may lead to a slightly elevated $R_{\text{ns}}$, suggesting a stronger impact of the attack on nonsource pixels. However, the influence of the dominance value $d$ on $R_{\text{ns}}$ is relatively insignificant, as $R_{\text{ns}}$ remains below 0.1% consistently. Consequently, selecting $d = 10$ proves advantageous for achieving optimal attack efficacy.

*3) Minimum Overlap Parameter for Training MSASS:* To facilitate the implementation of our MSASS, we strategically configure a sizable minimum overlap parameter, aiming to encompass multiple patches containing the source object within the training process. To assess the sensitivity of the proposed MSASS to the minimum overlap parameter, we conduct experiments with varying minimum overlaps (specifically, 100, 150, 200, and 250) while keeping the patch size fixed at 256 × 256. In these experiments, we employ U-Net and Unetformer as the pretrained victim models and utilize the Vaihingen dataset. The evaluation involves two source-to-target attack scenarios denoted as "Source class → Target class," with one

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TABLE X

ATTACK PERFORMANCE OF MSASS WITH DIFFERENT TRAIN MINIMUM OVERLAP IN THE VAIHINGEN SUMMER DATASET. FOR EACH ATTACK SCENARIO, THE SOURCE-TO-TARGET RATIO ($R_s$) AND OTHERS-TO-TARGET RATIO ($R_{\text{NS}}$) ARE REPORTED. THE HIGHER (LOWER) $R_s$ ($R_{\text{NS}}$), THE BETTER. ALL NUMBERS ARE REPORTED IN %

| Attack Scenario | Model | Train Overlap = 100 | | Train Overlap = 150 | | Train Overlap = 200 | | Train Overlap = 250 | |
|---|---|---|---|---|---|---|---|---|---|
| | | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ |
| Cars → Impervious surfaces | U-Net | 43.24 | 1.13 | 76.93 | 1.70 | 83.93 | 1.39 | 22.46 | 1.63 |
| | Unetformer | 48.76 | 1.14 | 49.14 | 0.79 | 75.66 | 0.91 | 58.42 | 0.72 |
| Buildings → Low vegetation | U-Net | 37.55 | 1.79 | 65.69 | 1.53 | 65.43 | 1.06 | 62.02 | 1.35 |
| | Unetformer | 39.56 | 1.32 | 60.80 | 1.71 | 90.25 | 1.85 | 55.64 | 1.56 |

TABLE XI

ATTACK PERFORMANCE OF "BUILDINGS → LOW VEGETATION" MSASS WITH DIFFERENT INFERENCE PATCH SIZES. FOR EACH SCENARIO, THE SOURCE-TO-TARGET RATIO ($R_s$) AND OTHERS-TO-TARGET RATIO ($R_{\text{NS}}$) ARE REPORTED. THE HIGHER (LOWER) $R_s$ ($R_{\text{NS}}$), THE BETTER. ALL NUMBERS ARE REPORTED IN %

| Inference Patch Size | Metric | Min Overlap = 0 | | Min Overlap = 100 | |
|---|---|---|---|---|---|
| | | U-Net | Unetformer | U-Net | Unetformer |
| $128 \times 128$ | $R_s(\uparrow)$ | 57.64 | 93.86 | 53.01 | 63.80 |
| | $R_{ns}(\downarrow)$ | 1.14 | 0.65 | 1.86 | 2.28 |
| $256 \times 256$ | $R_s(\uparrow)$ | 79.28 | 94.75 | 76.52 | 94.58 |
| | $R_{ns}(\downarrow)$ | 0.24 | 1.09 | 1.23 | 1.34 |
| $512 \times 512$ | $R_s(\uparrow)$ | 62.68 | 79.32 | 61.49 | 79.13 |
| | $R_{ns}(\downarrow)$ | 1.05 | 1.32 | 1.06 | 1.60 |

TABLE XII

SASS ATTACK PERFORMANCE IN A BLACK-BOX SETTING. THE SOURCE-TO-TARGET RATIO ($R_s$) AND OTHERS-TO-TARGET RATIO ($R_{\text{NS}}$) ARE REPORTED. THE HIGHER (LOWER) $R_s$ ($R_{\text{NS}}$), THE BETTER. ALL NUMBERS ARE REPORTED IN %

| Surrogate | Target | Zurich Summer | | Vaihingen | |
|---|---|---|---|---|---|
| | | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ |
| Unetformer | U-Net | 1.57 | 0.48 | 40.55 | 0.49 |
| U-Net | Unetformer | 26.24 | 1.12 | 37.62 | 1.01 |
| FCN8s | DeepLabV3+ | 2.68 | 0.13 | 49.78 | 0.19 |
| DeepLabV3+ | FCN8s | 3.57 | 0.06 | 10.59 | 0.11 |

victim source object chosen for each test image. To ensure fairness in the comparison, the same victim source object is used for each pretrained model. For equitable assessment under different minimum overlap parameters during training, we randomly select five patches containing the source object for testing. The summarized results in Table X present the source-to-target ratio ($R_s$) and others-to-target ratio ($R_{\text{ns}}$) for each attack scenario. Notably, as we increase the minimum overlap from 100 to 200, the attack performance demonstrates a consistent improvement. This can be attributed to the larger minimum overlap parameter, allowing the attack to leverage a greater number of patches, thereby enhancing its performance. However, upon exceeding a certain threshold, specifically at a training overlap of 250 in our setting, the attack performance experiences a slight decline. We postulate that the increased number of patches used for collectively training the perturbations may make optimization more challenging, potentially necessitating additional iterations for the objective function to converge. Considering the inherent variability in ground objects' spatial scales, we propose the customization of patch size and train minimum overlap to align with the specific characteristics of each object.

*4) Patch Size in the Inference Stage for MSASS:* To validate whether the proposed MSASS attack can succeed when the patch size used in the inference stage is different from the training stage, we further conduct experiments on the Vaihingen dataset. We fix the patch size at $256 \times 256$ pixels during the training stage and explore three different patch sizes in the inference stage: $128 \times 128$ pixels, $256 \times 256$ pixels, and $512 \times 512$ pixels. These represent cases where the training patch size is larger than, equal to, and smaller than the inference patch size. Notably, we directly use the same adversarial image generated during training without any further modification for inference, which means the only difference among these three cases lies in the inference patch size. Given the inference minimum patch size of $128 \times 128$, we assess the impact of inference minimum overlap at 0 and 100. We employ U-Net and Unetformer as the pretrained victim models and utilize the "Buildings → Low vegetation" attack scenario. The remaining experimental settings remain consistent with those employed previously. The outcomes are presented in Table XI.

The results consistently show effective attack performance across all three inference patch sizes with the lowest $R_s$ exceeding 53%. Optimal results were observed when the training patch size matched the inference patch size, as evidenced by the highest $R_s$. With an inference patch size of $128 \times 128$, the attack performance slightly decreases. However, it still achieves a notable stealthy attack effect, with $R_s$ at 93.86% and $R_{\text{ns}}$ at 0.65% on Unetformer, when the minimum overlap during inference is set to 0. When the inference patch size is increased to $512 \times 512$, a reduction in attack performance is also observed compared to the $256 \times 256$ scenario. Despite this decline, the attack still exhibits commendable efficacy, generally outperforming the $128 \times 128$ case. For instance, Unetformer achieves $R_s$ of 79.13% when the patch size is $512 \times 512$ and the minimum overlap is 100.

### F. Black-Box Analysis

In this study, we primarily focus on stealthy attacks in the white-box attack scenario for semantic segmentation. Here, we extend our analysis to discuss the applicability of these attacks in a black-box setting. We investigate whether adversarial examples generated from surrogate models retain the

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TABLE XIII
MSASS ATTACK PERFORMANCE IN A BLACK-BOX SETTING. THE SOURCE-TO-TARGET RATIO ($R_s$) AND OTHERS-TO-TARGET RATIO ($R_{\text{NS}}$) ARE REPORTED. THE HIGHER (LOWER) $R_s$ ($R_{\text{NS}}$), THE BETTER. ALL NUMBERS ARE REPORTED IN %

| Surrogate | Target | Zurich Summer | | Vaihingen | |
|---|---|---|---|---|---|
| | | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ | $R_s(\uparrow)$ | $R_{ns}(\downarrow)$ |
| Unetformer | U-Net | 0.56 | 0.92 | 35.74 | 4.19 |
| U-Net | Unetformer | 39.78 | 16.69 | 6.26 | 0.97 |
| FCN8s | DeepLabV3+ | 2.87 | 2.27 | 18.28 | 5.53 |
| DeepLabV3+ | FCN8s | 3.69 | 0.19 | 25.07 | 2.47 |

ability to effectively attack target models. The transferability of our proposed SASS and MSASS attacks is analyzed on the Vaihingen and Zurich Summer datasets, each featuring one attack scenario: "Cars → Impervious surfaces" for Vaihingen and "Bare Soil → Buildings" for Zurich Summer. For the MSASS attack, the test minimum overlap is 200. Two sets of models are chosen as surrogate and target models: U-Net and Unetformer, FCN8s, and DeepLabV3+. The experimental results are presented in Tables XII and XIII. Both SASS and MSASS attacks exhibit a certain degree of transferability in black-box settings. For instance, as shown in Table XII, the SASS perturbation generated by the surrogate model FCN8s achieves $R_s$ of 49.78% on the target DeepLabV3+ model. Similarly, in Table XIII, the SASS perturbation generated by the surrogate model U-Net achieves $R_s$ of 39.78% on the target Unetformer model. However, the overall performance in black-box settings is not uniform and is contingent upon the attack scenario and dataset. This variability suggests avenues for exploration in future research.

## V. CONCLUSION

In conclusion, this article introduced two novel methods, the SASS and its extension, the MSASS, aimed at enhancing the stealthiness and flexibility of adversarial attacks in remote sensing semantic segmentation. The SASS method selectively manipulates pixels from the source class to be classified as the target class while keeping nonsource classes unaffected, resulting in a more refined and targeted attack strategy. This approach demonstrates improved stealthiness and effectiveness in attacking remote sensing semantic segmentation models. Building upon SASS, the MSASS method addresses real-world scenarios where translation transformations may exist. It incorporates a masking technique to apply perturbations only to areas corresponding to the victim object and utilizes a minimum overlap scheme to effectively train the perturbation across multiple translation copies. This extension further enhances the attack's efficacy and adaptability in challenging environments. Extensive experiments were conducted to validate the effectiveness of both SASS and MSASS. The results showcased their strong attack performance in terms of stealthiness and overall effectiveness. In addition, a comparison between SASS and MSASS was performed, highlighting the advantages of the proposed extension. Furthermore, the impact of SASS and MSASS on the detection of spatial consistency was evaluated, demonstrating their ability to evade existing adversarial detection mechanisms. These findings underscore the significance of SASS and MSASS as powerful tools for enhancing the stealthiness and flexibility of adversarial attacks in remote sensing semantic segmentation. Overall, this study contributes valuable insights and methodologies to the field, paving the way for further advancements in the development of robust defense mechanisms against adversarial attacks in remote sensing applications.

## REFERENCES


[1] C. Xiao, R. Deng, B. Li, F. Yu, M. Liu, and D. Song, "Characterizing adversarial examples based on spatial consistency information for semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 217–234.

[2] X. Yuan, J. Shi, and L. Gu, "A review of deep learning methods for semantic segmentation of remote sensing imagery," *Expert Syst. Appl.*, vol. 169, May 2021, Art. no. 114417.

[3] F. Fang, X. Yuan, L. Wang, Y. Liu, and Z. Luo, "Urban land-use classification from photographs," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 12, pp. 1927–1931, Dec. 2018.

[4] J. K. Jadhav and R. P. Singh, "Automatic semantic segmentation and classification of remote sensing data for agriculture," *Math. Models Eng.*, vol. 4, no. 2, pp. 112–137, Jun. 2018.

[5] F. I. Diakogiannis, F. Waldner, P. Caccetta, and C. Wu, "ResUNet—A: A deep learning framework for semantic segmentation of remotely sensed data," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 94–114, Apr. 2020.

[6] H. Arief, G.-H. Strand, H. Tveite, and U. Indahl, "Land cover segmentation of airborne LiDAR data using stochastic atrous network," *Remote Sens.*, vol. 10, no. 6, p. 973, Jun. 2018.

[7] X. Pan, L. Gao, B. Zhang, F. Yang, and W. Liao, "High-resolution aerial imagery semantic labeling with dense pyramid network," *Sensors*, vol. 18, no. 11, p. 3774, Nov. 2018.

[8] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.

[9] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," 2017, *arXiv:1706.06083*.

[10] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2017, pp. 39–57.

[11] T. Bai, H. Wang, and B. Wen, "Targeted universal adversarial examples for remote sensing," *Remote Sens.*, vol. 14, no. 22, p. 5833, Nov. 2022.

[12] Y. Xu and P. Ghamisi, "Universal adversarial examples in remote sensing: Methodology and benchmark," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5619815.

[13] C. Xie, J. Wang, Z. Zhang, Y. Zhou, L. Xie, and A. Yuille, "Adversarial examples for semantic segmentation and object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, Jun. 2017, pp. 1–11.

[14] J. H. Metzen, M. C. Kumar, T. Brox, and V. Fischer, "Universal adversarial perturbations against semantic image segmentation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2755–2764.

[15] Z. Chen, C. Wang, and D. Crandall, "Semantically stealthy adversarial attacks against segmentation models," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2022, pp. 2846–2855.

[16] P. Benz, C. Zhang, T. Imtiaz, and I. S. Kweon, "Double targeted universal adversarial perturbations," in *Proc. Asian Conf. Comput. Vis.*, 2020, pp. 1–17.

[17] S. Paisitkriangkrai, J. Sherrah, P. Janney, and A. Van-Den Hengel, "Effective semantic pixel labelling with convolutional networks and conditional random fields," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2015, pp. 36–43.

[18] J. B. Campbell and R. H. Wynne, *Introduction to Remote Sensing*. New York, NY, USA: Taylor & Francis, 2011.

[19] J. Ma, W. Yu, C. Chen, P. Liang, X. Guo, and J. Jiang, "Pan-GAN: An unsupervised pan-sharpening method for remote sensing image fusion," *Inf. Fusion*, vol. 62, pp. 110–120, Oct. 2020.

[20] M. Zhao, P. Olsen, and R. Chandra, "Seeing through clouds in satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4704616.


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[21] J. Zhang, J. Lei, W. Xie, Z. Fang, Y. Li, and Q. Du, "SuperYOLO: Super resolution assisted object detection in multimodal remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605415.

[22] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[23] D. Rashkovetsky, F. Mauracher, M. Langer, and M. Schmitt, "Wildfire detection from multisensor satellite imagery using deep semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7001–7016, 2021.

[24] S. Dewitte, J. P. Cornelis, R. Müller, and A. Munteanu, "Artificial intelligence revolutionises weather forecast, climate monitoring and decadal prediction," *Remote Sens.*, vol. 13, no. 16, p. 3209, Aug. 2021.

[25] B. Janga, G. Asamani, Z. Sun, and N. Cristea, "A review of practical AI for remote sensing in Earth sciences," *Remote Sens.*, vol. 15, no. 16, p. 4112, Aug. 2023.

[26] M. Immitzer, C. Atzberger, and T. Koukal, "Tree species classification with random forest using very high spatial resolution 8-band WorldView-2 satellite data," *Remote Sens.*, vol. 4, no. 9, pp. 2661–2693, Sep. 2012.

[27] V. F. Rodriguez-Galiano, B. Ghimire, J. Rogan, M. Chica-Olmo, and J. P. Rigol-Sanchez, "An assessment of the effectiveness of a random forest classifier for land-cover classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 67, pp. 93–104, Jan. 2012.

[28] T. N. Phan, V. Kuch, and L. W. Lehnert, "Land cover classification using Google Earth Engine and random forest classifier—The role of image composition," *Remote Sens.*, vol. 12, no. 15, p. 2411, Jul. 2020.

[29] K. Yang et al., "High-resolution mapping of snow cover in montane meadows and forests using planet imagery and machine learning," *Frontiers Water*, vol. 5, Jun. 2023, Art. no. 1128758.

[30] M. Sheykhmousa, M. Mahdianpari, H. Ghanbari, F. Mohammadimanesh, P. Ghamisi, and S. Homayouni, "Support vector machine versus random forest for remote sensing image classification: A meta-analysis and systematic review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 6308–6325, 2020.

[31] J. Li and Z. Liu, "Efficient camera self-calibration method for remote sensing photogrammetry," *Opt. Exp.*, vol. 26, no. 11, p. 14213, 2018.

[32] L. Wen, Y. Cheng, Y. Fang, and X. Li, "A comprehensive survey of oriented object detection in remote sensing images," *Expert Syst. Appl.*, vol. 224, Aug. 2023, Art. no. 119960.

[33] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.

[34] Y. Xu, T. Bai, W. Yu, S. Chang, P. M. Atkinson, and P. Ghamisi, "AI security for geoscience and remote sensing: Challenges and future trends," *IEEE Geosci. Remote Sens. Mag.*, vol. 11, no. 2, pp. 60–85, Jun. 2023.

[35] W. Czaja, N. Fendley, M. Pekala, C. Ratto, and I.-J. Wang, "Adversarial examples in remote sensing," in *Proc. 26th ACM SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, Nov. 2018, pp. 408–411.

[36] L. Chen, Z. Xu, Q. Li, J. Peng, S. Wang, and H. Li, "An empirical study of adversarial examples on remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7419–7433, Sep. 2021.

[37] M. Lu, Q. Li, L. Chen, and H. Li, "Scale-adaptive adversarial patch attack for remote sensing image aircraft detection," *Remote Sens.*, vol. 13, no. 20, p. 4078, Oct. 2021.

[38] Y. Zhang et al., "Adversarial patch attack on multi-scale object detection for UAV remote sensing images," *Remote Sens.*, vol. 14, no. 21, p. 5298, Oct. 2022.

[39] X. Sun, G. Cheng, L. Pei, H. Li, and J. Han, "Threatening patch attacks on object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5609210.

[40] Y. Na, J. H. Kim, K. Lee, J. Park, J. Y. Hwang, and J. P. Choi, "Domain adaptive transfer attack-based segmentation networks for building extraction from aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5171–5182, Jun. 2021.

[41] Y. Mo, Y. Wu, X. Yang, F. Liu, and Y. Liao, "Review the state-of-the-art technologies of semantic segmentation based on deep learning," *Neurocomputing*, vol. 493, pp. 626–646, Jul. 2022.

[42] S.-A. Boukabara, J. Eyre, R. A. Anthes, K. Holmlund, K. M. S. Germain, and R. N. Hoffman, "The Earth-observing satellite constellation: A review from a meteorological perspective of a complex, interconnected global system with extensive applications," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 3, pp. 26–42, Sep. 2021.

[43] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[44] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, vol. 9351, 2015, pp. 234–241.

[45] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[46] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[47] H. Zhao, X. Qi, X. Shen, J. Shi, and J. Jia, "ICNet for real-time semantic segmentation on high-resolution images," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 405–420.

[48] X. He et al., "Semantic segmentation of remote-sensing images based on multiscale feature fusion and attention refinement," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[49] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[50] J. Gu, H. Zhao, V. Tresp, and P. H. Torr, "SegPGD: An effective and efficient adversarial attack for evaluating and boosting segmentation robustness," in *Proc. 17th Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 308–325.

[51] Z. Wang, B. Wang, Y. Liu, and J. Guo, "Global feature attention network: Addressing the threat of adversarial attack for aerial image semantic segmentation," *Remote Sens.*, vol. 15, no. 5, p. 1325, Feb. 2023.

[52] J. Lu, H. Sibai, E. Fabry, and D. Forsyth, "NO need to worry about adversarial examples in object detection in autonomous vehicles," 2017, *arXiv:1707.03501*.

[53] A. Arnab, O. Miksik, and P. H. S. Torr, "On the robustness of semantic segmentation models to adversarial attacks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 888–897.

[54] A. Fawzi and P. Frossard, "Manitest: Are classifiers really invariant?" 2015, *arXiv:1507.06535*.

[55] Z. Zhao, Z. Liu, and M. Larson, "On success and simplicity: A second look at transferable targeted attacks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 6115–6128.

[56] X. Sun, G. Cheng, H. Li, L. Pei, and J. Han, "On single-model transferable targeted attacks: A closer look at decision-level optimization," *IEEE Trans. Image Process.*, vol. 32, pp. 2972–2984, 2023.

[57] M. Cramer, "The DGPF-test on digital airborne camera evaluation overview and test design," *Photogrammetrie-Fernerkundung-Geoinf.*, vol. 2010, no. 2, pp. 73–82, May 2010.

[58] M. Volpi and V. Ferrari, "Semantic segmentation of urban scenes by learning local class interactions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2015, pp. 1–9.

[59] M. Li, C. Deng, T. Li, J. Yan, X. Gao, and H. Huang, "Towards transferable targeted attack," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 641–649.



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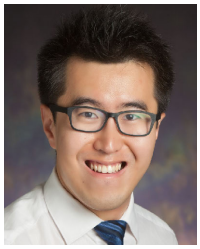
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